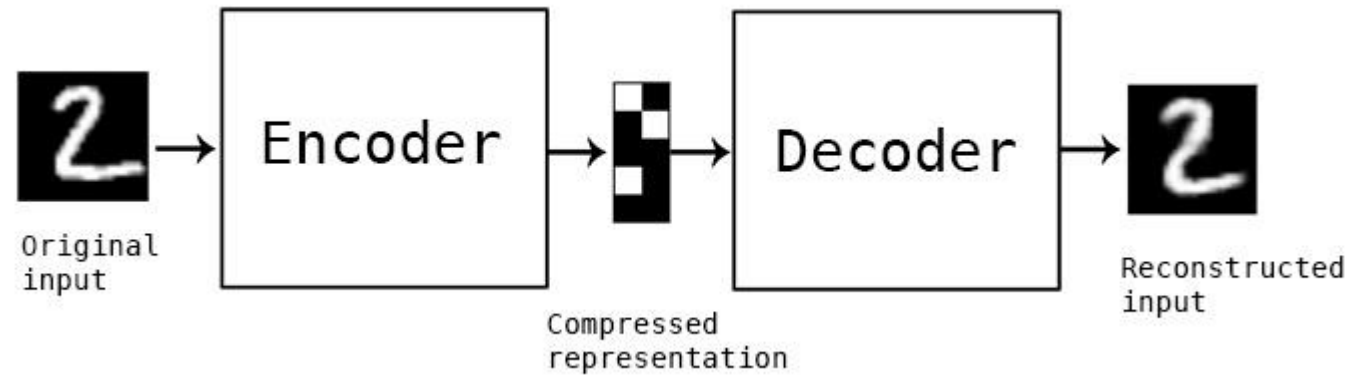


COMP4616 – Fundamentals of Machine Learning

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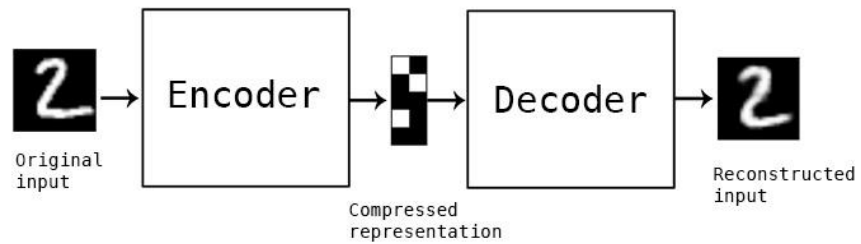
SPRING 2026

TYPES OF ANN - PART 2: AUTOENCODERS (AE), GENERATIVE ADVERSARIAL NETWORKS (GAN)



Autoencoders

What is an Autoencoder?



- **Autoencoders** learn to compress and reconstruct input data without labels (unsupervised learning)

Objective

- Minimize reconstruction error between input and output (e.g. MSE loss).

Two Part Architecture

Encoder

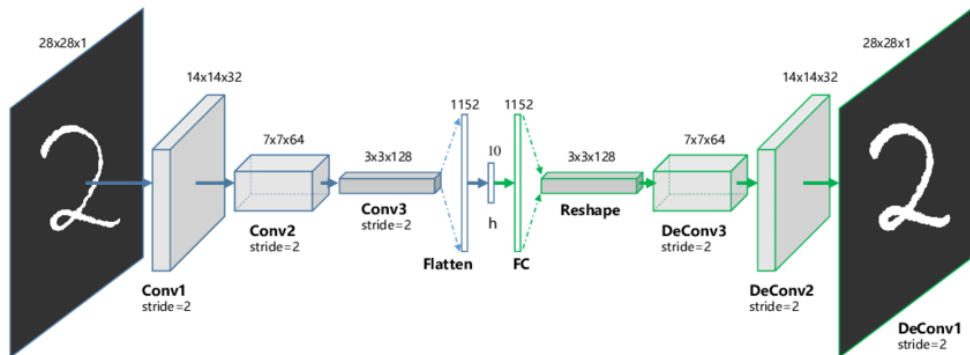
- Compresses input to a lower-dimensional latent space. Captures the essential structure of the data.

Latent Space (Bottleneck)

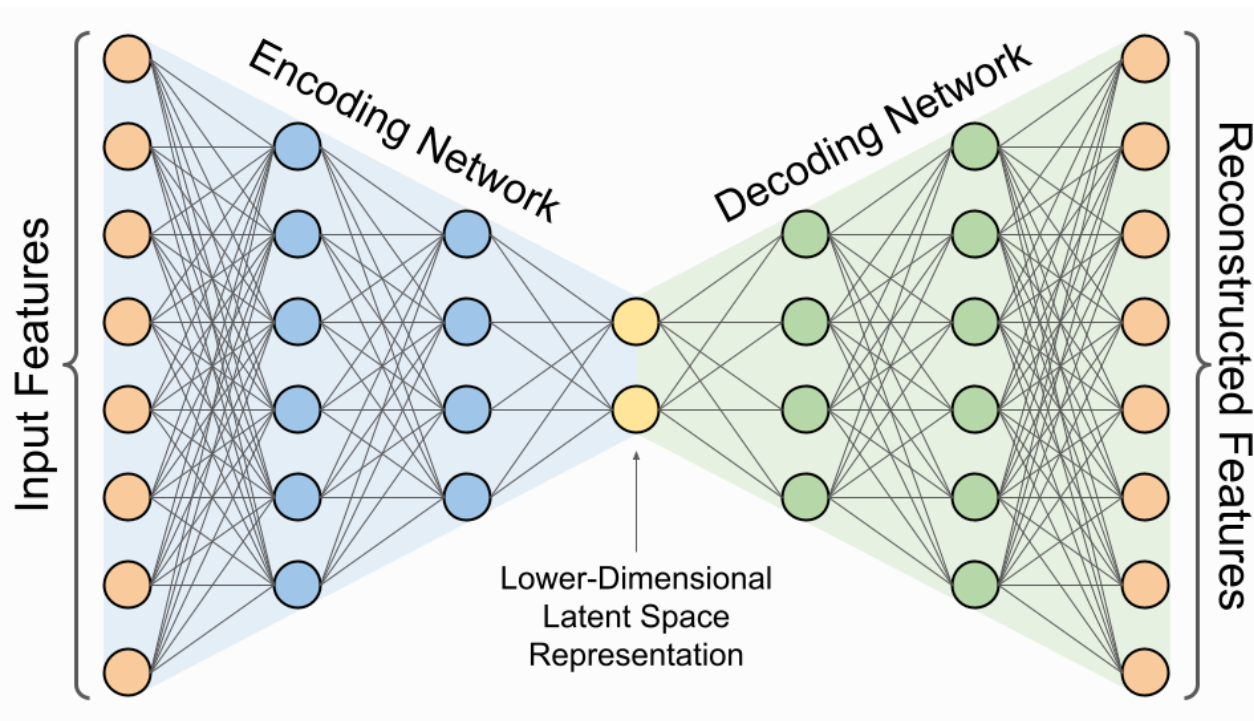
- The compressed representation h . Forced to capture only the most important features.

Decoder

- Aims for minimal reconstruction error.



Autoencoder Architecture



Encoding Network

- Compresses input through multiple layers
- Each layer reduces dimensionality, forcing the network to learn efficient representations

Lower Dimensional Latent Space

- The bottleneck layer h
- Dimensionality chosen by design
- In VAEs, this is a probability distribution

Decoding Network

- Mirror of the encoder
- Progressively expands back to the original input dimensions
- Trained jointly with the encoder

Autoencoder Applications

Dimensionality Reduction

- Autoencoders can reduce the dimensionality of data
- Nonlinear alternative to PCA but captures complex data structure

Denoising

- Remove noise by training on corrupted → clean pairs

Anomaly Detection

- High reconstruction error signals an anomaly

Generative Models

- Variational AEs generate new samples similar to training data.

Autoencoder Trade offs

Advantages

- Unsupervised, no labels needed
- Learns meaningful feature representations
- Identifies anomalies via reconstruction error

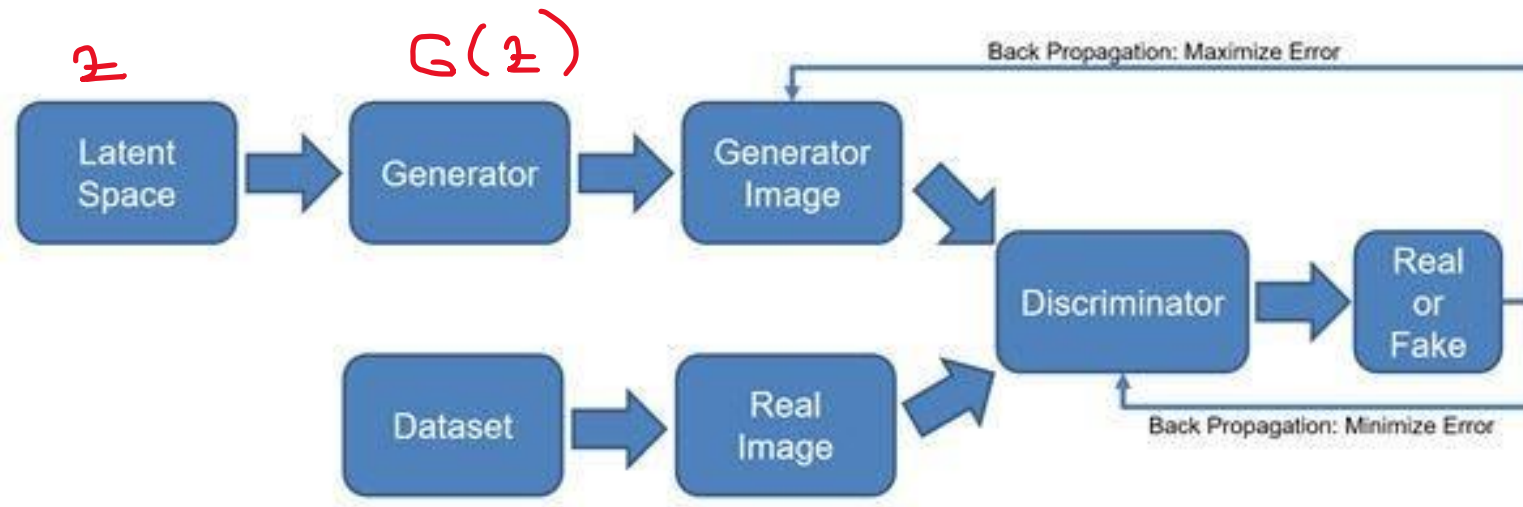
Challenges

- Prone to overfitting with large capacity
- Information loss with small latent space
- Autoencoders are handcrafted for specific data distributions → Limited generalization to unseen data distributions

Generative Adversarial Networks

Generative Adversarial Networks (GAN)

- A **Generative Adversarial Network (GAN)** is a type of neural network architecture that consists of two models, a **generator** and a **discriminator**, which are **trained simultaneously** through a process of **adversarial competition**.
- GANs were introduced by Ian Goodfellow and his colleagues in 2014.



GAN Architecture

- **Generator:**

- The generator takes random noise as input and **generates synthetic data** (e.g., images) that **aims to resemble real data**.
- Its goal is to produce data that is **indistinguishable** from the real data.

- **Discriminator:**

- The discriminator takes both real data and synthetic data produced by the generator as input and tries to **distinguish** between them.
- It outputs a probability indicating whether the input data is **real** or **fake**.

GAN Training Process

Sample real & fake data

- A batch of real data from the dataset and a batch of fake data generated by $G(z)$ are collected.

Train the Discriminator

- D is updated to better distinguish real from fake by maximizing its classification accuracy

Train the Generator

- G is updated to fool D
- G generates data that D incorrectly classifies as real
- G's loss = D's confidence in fake data being real

Zero-Sum Game

- Training converges when G produces data indistinguishable from real, and D outputs 0.5 for every sample.

Applications of GANs

- **Image Generation**

- GANs can generate high quality images from random noise.
- Examples include generating realistic photographs, art, and faces (e.g., DeepArt, DeepFake).

- **Image to Image Translation**

- GANs can be used for tasks like converting sketches to photos, black and white images to color, or daytime scenes to nighttime scenes (e.g., pix2pix, CycleGAN).

- **Text to Image Synthesis**

- GANs can generate images from textual descriptions (e.g., generating an image of a bird based on a textual description of its features by using Stable diffusion, Adobe Firefly).

- **Super Resolution**

- GANs can enhance the resolution of images, generating high resolution images from low resolution inputs. (e.g. [SRGAN](#))

- **Data Augmentation**

- GANs can generate synthetic data to augment training datasets, improving the performance of machine learning models. (e.g. [Motamed et. al.](#))

Mode collapse

What is mode collapse?

- The generator learns to produce a limited variety of outputs instead of the full data distribution

Example

- A GAN trained on MNIST digits only generates "3"s regardless of input noise

Why does mode collapse occur?

- Adversarial instability, generator overfitting, or imbalanced G/D training

Pros & Cons of GANs

- **Advantages:**

- Ability to generate high quality synthetic data
- Useful in various applications, including image generation, data augmentation, and style transfer

- **Challenges**

- Training GANs can be difficult due to instability and **mode collapse**.
- Requires careful tuning of the training process and network architecture